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Quantum Fuzzy Neural Network for multimodal sentiment and sarcasm detection

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ABSTRACT

Sentiment and sarcasm detection in social media contribute to assessing social opinion trends. Over the years, most artificial intelligence (AI) methods have relied on real values to characterize the sentimental and sarcastic features in language. These methods often overlook the complexity and uncertainty of sentimental and sarcastic elements in human language. Therefore, this paper proposes the Quantum Fuzzy Neural Network (QFNN), a multimodal fusion and multitask learning algorithm with a Seq2Seq structure that combines Classical and Quantum Neural Networks (QNN), and fuzzy logic. Complex numbers are used in the Fuzzifier to capture sentiment and sarcasm features, and QNN are used in the Defuzzifier to obtain the prediction. The experiments are conducted on classical computers by constructing quantum circuits in a simulated noisy environment. The results show that QFNN can outperform several recent methods in sarcasm and sentiment detection task on two datasets (Mustard and Memotion). Moreover, by assessing the fidelity of quantum circuits in a noisy environment, QFNN was found to have excellent robustness. The QFNN circuit also possesses expressible and entanglement capabilities, proving effective in various settings. Our code is available at https://github.com/prayagtiwari/QFNN.

1. Introduction

In recent years on social media, people are accustomed to utilizing multimodal information to convey their feelings [1]. For instance, in response to current trending events, individuals can express their views through text and some memo images or using short videos. In this context, the identification of sarcasm and sentiment features plays an important role for judging the current trend of societal opinion. By analyzing sarcasm and sentiment tendencies in individuals language, we can understand the public's view of current social events. Following the implementation of any governmental policy, the government can infer the public sentiment and sarcasm expressed on social media and then understand whether this policy is worth promoting. On the other hand, when companies conduct market research for products, they can also understand the market satisfaction of the product by conducting a unified sarcasm and sentiment analysis of user product evaluations. This approach can save substantial manpower costs. There are many such applicable scenarios, and therefore, multimodal sarcasm and sentiment analysis has received a lot of attention in recent years [2].

Detecting multomodal sarcasm and sentiment involves extracting sentiment and sarcasm features by the aggregation of context features and the fusion of multimodal features. On the other hand, in multitasking learning, where sentiment detection is considered a primary task and sarcasm is the secondary task, one can learn the correlations from sarcasm to efficiently detect sentiment. This detection method distinguishes itself from single-modal and single-task methods, leveraging information from multiple modalities to capture sarcasm and sentiment features more comprehensively. It uses one task's features in multimodal settings to enhance another task's ability based on some correlations.

The commonly used methods for the multimodal sarcasm and sentiment detection are as follows: For the aggregation of information context, commonly used methods include RNN and its variants LSTM,

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GRU, etc. Attention mechanisms are also used to analyze the impact of each context. There are currently methods such as the conversational memory network (CMN) [3], dialogue recurrent neural network (DialogueRNN) [4], dialogue graph convolutional network (DialogueGCN) [5], quantum-inspired interactive network (QIN) [6] and iterative sentiment interaction network (IEIN) [7]. In multimodal fusion, there are currently three levels, including feature-level fusion (early fusion), decision-level fusion (late fusion), and their combination (hybrid fusion) [8]. The goal of early fusion is to capture the correlation between multimodal features. Its main obstacle arises from information heterogeneity since each modal feature comes from different semantic spaces. In contrast, late fusion aims to combine based on local decisions of individual features. This method is more flexible than early fusion but ignores the correlation between multimodal features. The hybrid fusion method fully utilizes the advantages of early and late fusion strategies, which is the multimodal fusion method adopted in this paper. For the multitask learning of sarcasm and sentiment, it should be noted that the features of one task can enhance the expression ability of another task's features. But if one task's detection affects too much in the detection of another one, it will also reduce the detection accuracy of the affected task. To mitigate this issue, Masked-Attention was used to reduce the number of shared parameters between the two tasks to prevent excessive intervention.

However, the current multimodal multitask sentiment and sarcasm detection method still has problems with insufficient expression of sentiment and sarcasm features. Classical neural networks currently have no effective way to fully express the complexity of sentiment and sarcasm parts in human language.

In addressing this issue, this paper introduces Quantum Fuzzy Neural Network (QFNN), a novel algorithm designed for multimodal fusion and multitask learning. The proposed QFNN utilizes a hybrid classical and Quantum Neural Network, enabling efficient context aggregation and multimodal fusion. It enhances the representation capacity of sarcasm and sentiment features, leading to improved performance in sarcasm and sentiment detection.

1.1. Contribution

- We constructed the QFNN, which enhances the expression of sarcasm and sentiment features and increases the performance of sarcasm and sentiment detection.
- We designed the Quantum Fuzzy Composition consist of Fuzzifier and Defuzzifier separately to realize the fuzzy expression of sentiment and sarcasm features on quantum circuits.
- We conducted experiments on the Mustard and Memotion datasets to evaluate the performance of our proposed model. QFNN's performance is compared against several models, and found to be effective. Additionally, we compared the performance of the model under different modality information and task order.
- Furthermore, we evaluated the performance of QFNN in various aspects, including noise resistance, expressibility and entanglement capability, and also compared the impact of different quantum encoding methods.

1.2. Organization

The organization of this article is as follows: Section 2 provides an overview of the related work, discussing the current state of research on sarcasm and sentiment detection, as well as research related to QFNN. Section 3 presents the main algorithm and the implementation details of QFNN. Section 4 presents experimental results to show the performance and robustness of the algorithm. Finally, Section 5 summarizes our work with the conclusion and offers insights into future directions for research.

2. Related work

2.1. Sarcasm and sentiment detection

2.1.1. Multimodal fusion

In recent years, the sarcasm and sentiment detection research community has focused on multimodal information fusion for analysis. Specifically, various methods have been explored to improve multimodal information fusion and enhance language expression for these tasks.

For sarcasm detection, the initial approaches were *rule-based*. In 2010, Veale and Hao [9] followed by Riloff et al. (2013) [10], Maynard and Greenwood (2014) [11], and Bharti et al. (2015) [12], utilized *rule-based* approaches. These methods relied on hand-crafted features, which required manual preparation of all features before training, resulting in high algorithmic costs. To address the limitations of rule-based approaches, Deep Learning methods emerged as a solution. These methods automatically extract features during training, significantly improving algorithm efficiency. In 2016, Schifanella et al. [13] introduced a multimodal deep learning method to fuse text and image information. However, the neural network's structure was relatively simple and could not fully capture the complexity of sarcasm features. In 2019, Cai et al. [14] utilized a hierarchical fusion method for multimodal information in neural networks, enhancing the model's expressive capability. However, their approach did not consider audio information. In 2020, Srivastava [15] employed BERT to construct a hierarchical structure, but it lacked the ability to process large text information effectively. In 2021, Bedi et al. [16] addressed non-English sarcasm detection by utilizing numerous attention mechanisms in neural network structures based on the Hindi-English database.

For sentiment analysis, the multimodal deep learning method still offer advantages. In 2017, Zadeh et al. [17] used tensor products to fuse multimodal information. They did not consider the aggregation of context and did not conduct thorough analysis of the extent of the influence between different modalities. This limited their ability to fully uncover the hidden sentiment features. Therefore, in 2019, Xu et al. [18] used a multimodal detection model based on the attention mechanism to analyze the influence between modalities. In 2017, Cambria et al. [19] used the existing methods in sentiment analysis tasks. In 2019, Poria and Cambria [2] explained in more detail the state-of-the-art approaches, including unimodality and multimodality.

2.1.2. Multitask learning

Several works show that one detection task's features can help another task's detection. In 2019, Majumder et al. [20] proposed a multimodal multitask method, showing that sarcasm detection can help sentiment detection tasks. In 2020, Pan et al. [21] proposed a multitask detection model for sarcasm and sentiment, but did not consider the aggregation of context. In 2023, Zhang et al. [22] constructed a multitask multimodal context-inclusive sarcasm and sentiment detection model, but they did not consider enhancing the ability to express sarcasm and sentiment features.

In summary, up to now, the development of sarcasm and sentiment detection has many cases in terms of multimodal information fusion, context aggregation, and multitask learning. However, there is no related research on the effective representation learning of sentiment and sarcasm features.

2.2. Quantum Machine Learning

Quantum Machine Learning (QML) is an emerging research area in AI. QML encompasses concepts such as Quantum Neural Networks (QNN), quantum kernel methods, Quantum Support Vector Machine (QSVM), and others. In recent years, QML community has made efforts to develp a variety of QNN models. QNNs use Parameterized Quantum Circuits (PQCs) to process input data. Unlike traditional neural

networks, QNNs leverage quantum entanglement and superposition for better information representation.

Kak et al. [23] first proposed the quantum neural computation concept in 1995, which provides a theoretical basis for QNN. Since then, various QML algorithms have emerged. For instance, in 2013, Lloyd et al. [24] presented a quantum K-means algorithm that is more efficient than its classical counterpart. In 2014, Wiebe et al. [25] used quantum states to represent classical vectors and introduced the quantum nearest neighbor algorithm. In the same year, Rebentrost et al. [26] used density matrices in quantum systems to describe the kernel method in support vector machines. Later, Wiebe et al. [27] suggested integrating QNNs with classical neural networks to boost training speed. However, many of the aforementioned studies have theoretical characteristics, and several algorithms cannot be implemented effeciently on the avaialble quntum resources. In the noisy intermediate scale quantum era, Variational Quantum Algorithms (VQA) have become an important research direction, with QNN based on quantum circuits receiving widespread attention. In 2018, Farhi et al. [28] developed QNN for binary classification and optimized its parameters. Schuld et al. [29] later updated QNN structure for better efficiency. In 2019, Cong et al. [30] proposed Quantum Convolution Neural Network (QCNN) to process classical image data. By constructing reasonable quantum circuits, Pesah et al. [31] et al. proved that QNN would not suffer from the barren plateau phenomenon, demonstrating the potential advantages. In 2021, Y. Du et al. [32] successfully used a QNN for a classification task with depolarizing noise, proving QNNs' robustness on quantum noise. In 2022, Qu et al. [33] proposed a spatio-temporal Quantum Graph Convolution Neural Network to address the traffic congestion prediction (TCP) problem. The results show that the quantum algorithm can effectively solve the TCP problem and has good resistance to interference. Later, QNN was applid in the smart healthcare field [34], where QCNN was used for feature extraction and Variational Quantum Classifier circuits for intelligent diagnosis. Also, in 2023, S. Mensa et al. [35] combined the classical support vector classifier algorithm with Ligand-Based Virtual Screening quantum kernel estimation on real databases to achieve a potential quantum advantage. The authors demonstrated that, under certain circumstances, quantum algorithms can provide advantages over most classical algorithms.

In general, QML has a mature theoretical basis after many years of development and has broad application prospects. However, the existing QNN achievements have not been applied to sarcasm detection and sentiment detection. At the same time, since quantum fuzzy logic is a relatively new field, most of work did not consider using Fuzzy Logic in QNN to represent features.

2.3. Quantum Fuzzy logic

Not all problems can be solved solely through classical Boolean logic [36]. Sometimes, having only two values is insufficient to solve the problem. In 1965, Zadeh proposed Fuzzy Logic [37], using a broader representation of variables, Fuzzy Logic can handle the diverse expressions of complex problems [38]. Quantum Fuzzy Logic can be connected from the perspective of Quantum Theory and Fuzzy Logic. In 2007, Melnichenko and Grigorii [39] mentioned that Quantum Logic is also a kind of Fuzzy Logic, and provided related mathematical proofs. This can serve as a theoretical basis for using fuzzy logic in quantum physics. In 2021, Gentili and Luigi [40] proposed a relation between fuzzy logic, neuroscience, and quantum theory through Bayesian probability that may inspire new ideas for developing AI models. Quantum fuzzy logic has been a less explored area and it could be impactful in different applications.

However, many models are proposed based on traditional fuzzy logic. In 2018, Kropat et al. proposed time-discrete target-environment regulatory systems [41]. In order to model the relationship between target and environmental entities in regulatory networks, they proposed a fuzzy target-environment network and analyzed the fuzziness of the regulatory system. Then, they extended the concept of fuzzy target-environment networks to fuzzy regression models with fuzzy data sets in order to understand interconnected complex systems. From 2020 to 2021, Goli et al. proposed a series of fuzzy models to solve issues such as the cell formation problem [42], relief vehicles problem [43], transportation route planning [44], etc. The fuzzy logic-based models were not applied in the field of sarcasm and sentiment detection. Therefore, in 2021, Zhang et al. [45] proposed a quantum-inspired complex-valued fuzzy sarcasm detection algorithm. They used state vectors similar to those in quantum information theory to express features and used membership functions in fuzzy logic to characterize the weights of states. However, their algorithm is still within the scope of classical neural networks and does not utilize QNN based on PQCs.

3. Methodology

The QFNN is composed of Encoder, Decoder, and Quantum Fuzzy Composition. The archiecture of QFNN is shown in Fig. 1.

3.1. Problem formulation

The original information from the three modalities is divided into target features and context features through feature extraction. The target features of the text is set as T_i^t, and the context features as T_j^c; the target features of the image as V_i^t, and the context features as V_j^c; the target features of the audio as A_i^t, and the context features as A_j^c. Here, $i \in \{0, 1, \dots N-1\}$, N represents the number of samples; $j \in \{0, 1, \dots n-1\}$, n represents the number of contexts. The goal of the algorithm is to predict the Sarcasm Label L_i^{sar} and Sentiment Label L_i^{sen}.

3.2. Proposed algorithms

3.2.1. Encoder

The primary function of the Encoder is to aggregate the context features of the single modality using Intra-modal Attention and fuse the features of different modalities using Inter-modal Attention. The following components are explained separately.

Intra-modal Attention: Intra-modal Attention is used to aggregate the context features of the same modality. Different context features have different impacts on the target features. Intra-modal Attention can discover context features with a larger impact.

Let $U \in \{T, V, A\}$, U_i^t be the target feature, and U_j^c be the context feature under this modality. W is a trainable parameter. The attention mechanism is first used to calculate the weights of each context feature for the target feature. The computation process is given as follows:

$$E_{i,j} = \tanh\left(W_i\left[U_i^t; U_j^c\right]\right) \tag{1}$$

$$\alpha_{i,j} = \mathrm{Softmax}\left(w^T E_{i,j}\right) \tag{2}$$

$$c = \sum_{j=1}^{n} \alpha_{i,j} U_j^c \tag{3}$$

Eq. (1) and (2) are the weight calculation operations in the attention mechanism. $E_{i,j}$ is the attention score between jth context feature and the target feature. $\alpha_{i,j}$ is the weight of the impact of this context feature on the target feature. c is the weighted representation of the context's impact on the target feature. Let M^U be the aggregation feature of context features and the target feature of modality U. W_s and W_p are the trainable weights. Then the residual connection is used to merge c and the target feature U_i^t:

$$M^U = U_i^t + \tanh\left(W_s c_a + W_p U_i^t\right) \tag{4}$$

The residual connection in (4) is used to mitigate the vanishing gradient problem, improve the training, and improve the efficiency of representation learning.

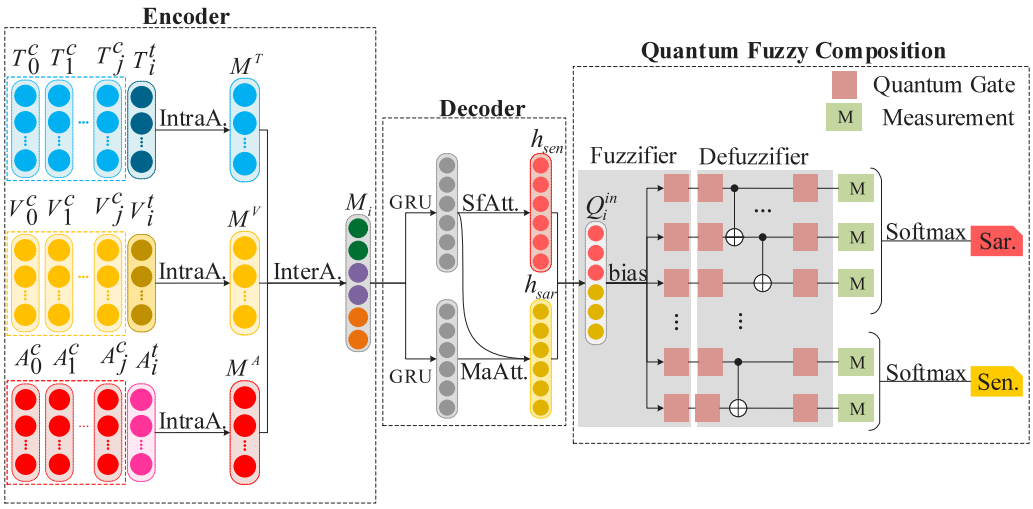


Fig. 1. Structure of QFNN: IntraA denotes Intra-modal Attention, InterA represents Inter-modal Attention, SfAtt represents Self Attention, Sar indicates Sarcasm Label, Sen indicates Sentiment Label.

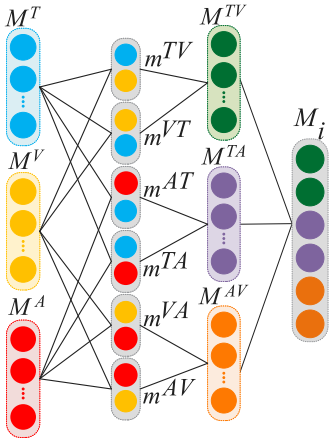


Fig. 2. Structure of Inter-modal Attention.

Inter-modal Attention: Inter-modal Attention is used to fuse features of different modalities. Modalities influence each other, and Inter-modal Attention can discover the degree of influence between different modalities.

As shown in Fig. 2, let $L_1 \in \{T, V, A\}, L_2 \in \{T, V, A\}, L_1 \neq L_2$. And $m^{L_1L_2}$ represents the attention weight representation of modality L_1 and modality L_2. We have:

$$E_{L_1,L_2} = tanh(W[M^{L_1}; M^{L_2}]) \tag{5}$$

$$\alpha_{L_1,L_2} = Softmax(w^T E_{L_1,L_2}) \tag{6}$$

$$m^{L_1L_2} = \alpha_{L_1,L_2} M^{L_1} \tag{7}$$

Eq. (5) and (6) calculate the attention score E_{L_1,L_2} and weight α_{L_1,L_2}, respectively. And $m^{L_1L_2}$ is the weight representation of attention. Let $M^{L_1L_2}$ be the fusion of modalities L_1 and L_2. We have:

$$M^{L_1L_2} = concat(m^{L_1L_2}, m^{L_2L_1}) \tag{8}$$

$M^{L_1L_2}$ in (8) is the aggregation of two modalities, where *concat* represents the concatenation of two features. Let M_i be the fusion of multiple modalities. We have:

$$M_i = concat(M^{TV}, M^{TA}, M^{AV}) \tag{9}$$

From (9), the fusion features of three models M_i are obtained.

3.2.2. Decoder

The Decoder function decodes the fusion features into sarcasm and sentiment features. Self-attention mechanisms and Gated Recurrent Unit (GRU) are used to distinguish between the two features. Masked attention is then used to represent the relationship between the two features. GRU is a type of Recurrent Neural Network (RNN), proposed to solve problems such as long-term memory and gradients in back-propagation. GRU is used to mine the hidden features in the fused features:

$$I_{sen} = GRU(M_i) \tag{10}$$

$$I_{sar} = GRU(M_i) \tag{11}$$

In Eq. (10) and (11), the hidden sarcasm and hidden sentiment features are denoted as I_{sen} and I_{sar}. Sentiment features can be helpful for the detection of sarcasm features. Therefore, Masked-attention is used to incorporate some hidden sentiment features into sarcasm features. The reason for using Masked-attention is to reduce the shared parameters in the two tasks of sarcasm and sentiment detection. For sentiment features, the self-attention mechanism is used to extract important parts from the features:

$$E_{sen} = tanh(W[I_{sen}; I_{sen}]) \tag{12}$$

$$\alpha_{sen} = Softmax(w^T E_{sen}) \tag{13}$$

$$h_{sen} = \alpha_{sen} I_{sen} \tag{14}$$

The above three equations represent the process of the self-attention mechanism. In Eq. (12), E_{sen} is the attention score. In Eq. (13), α_{sen} is the attention weight. In Eq. (14), h_{sen} is the weighted representation of the sentiment feature. Let t_{sen} be the hidden feature of M_i in GRU in Eq. (10). So, we have:

$$I_c = concat(I_{sar}, t_{sen}) \tag{15}$$

$$E_{sar} = tanh(W[I_c; I_c]) \tag{16}$$

$$\alpha_{sar} = Softmax(w^T E_{sar}) \tag{17}$$

$$h_{sar} = \alpha_{sen} I_c \tag{18}$$

In Eq. (15), the sarcasm feature and the hidden sentiment feature are first concatenated to get I_c. This step is to establish the connection between sentiment detection task and sarcasm detection task. Use hidden features in sentiment features to increase the expressiveness of sarcasm features. Eqs. (16), (17), (18) apply the self-attention mechanism to I_c to obtain the weighted sarcasm feature h_{sar}. In this step, the sarcasm detection task and sentiment detection task are connected. In

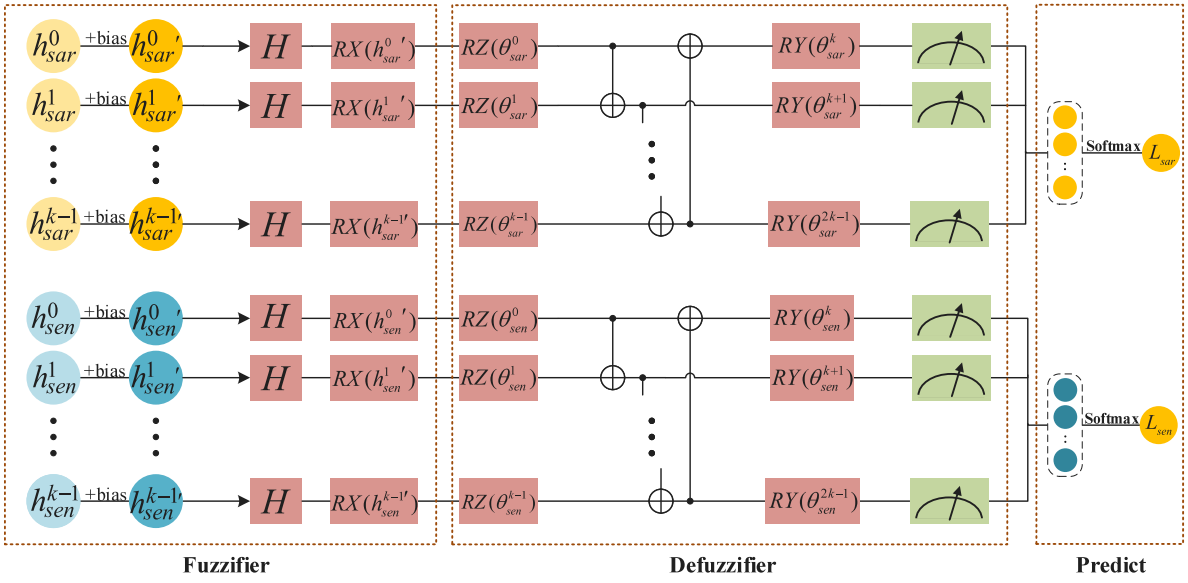


Fig. 3. Structure of Quantum Fuzzy Composition.

the following sections, more attention is paid to expressing features of QFNN using Quantum Fuzzy Composition. This is also supplemented with the detailed analysis in the experimental section.

3.2.3. Quantum Fuzzy composition

As shown in Fig. 3, the Quantum Fuzzy Composition of QFNN includes a Fuzzifier and Defuzzifier. The algorithmic details are discussed below.

Quantum Fuzzy Sets: Fuzzy sets were introduced and described using membership functions by L.A. Zadeh in 1965 [37]. The classical fuzzy sets are defined as:

$$A = \{(x, \mu_A(x)) \mid x \in \mathbb{X}\} \tag{19}$$

Here, x is the numerical value of a specific object, and X is its domain. $\mu_A(x)$ is the membership function of the object x. The membership function maps the fuzzy representation of the object. However, in quantum circuits, this representation needs to be slightly modified. Let $|\psi\rangle$ be a single qubit at a certain moment in the quantum circuit, where:

$$|\psi\rangle = p_1|\psi\rangle_1 + p_2|\psi\rangle_2 \cdots + p_n|\psi\rangle_n \tag{20}$$

Then, Quantum Fuzzy Sets can be represented as:

$$|\psi\rangle = \{(|\psi\rangle_i, p_i^2) \mid i \in [1, n]\} \tag{21}$$

Where,

$$\sum_i p_i^2 = 1 \tag{22}$$

From the above formula, it can be seen that Quantum Fuzzy Sets fit very well with the original expression of quantum states. That is to say, the state of a single qubit at each moment can be regarded as a fuzzy set.

Quantum Fuzzy System: A typical fuzzy system consists of a Fuzzifier and a Defuzzifier. This paper builds a Quantum Fuzzy System based on this structure. For the Fuzzifier, this paper is inspired by [45] and uses complex values to fuzzify sarcasm and sentiment features. For the Defuzzifier, this paper designs a parameterized quantum circuit to transform the fuzzy set into the corresponding category of quantum states.

Fuzzifier: In terms of language expressiveness, complex value features can express more information than real value features. This complex-valued representation is also a fuzzy representation of information [45]. Quantum circuits are naturally compatible with complex value information so that complex value information can be fully utilized.

Let the sentiment feature obtained through the Decoder be $h_{sen} = (h^0_{sen}, h^1_{sen}, \ldots, h^{k-1}_{sen})$, and the sarcasm feature be $h_{sar} = (h^0_{sar}, h^1_{sar}, \cdots h^{k-1}_{sar})$, where k is the length of the two features. And adding a bias to the features to obtain:

$$h^{t'}_{sen} = h^t_{sen} + b^t_{sen} \tag{23}$$

$$h^{t'}_{sar} = h^t_{sar} + b^t_{sar} \tag{24}$$

In the above, b is the bias, and t represents the tth attribute of the feature. b is a trainable parameter in this network. This step involves introducing a global phase after feeding the features into the quantum circuit. This global phase will then be a part of the complex encoding. It is worth noting that, during experiments, it was found that although a global phase can also be achieved by directly adding a parameterized rotation gate to the quantum circuit, it reduces the training efficiency.

Suppose the quantum circuit has $2k$ qubits, where the first k qubits are used for sarcasm feature detection, and the last k qubits are used for sentiment feature detection. The initial state of each qubit is $|0\rangle$. We use the Rx gate to encode the feature vector, which is called angle encoding. Taking the encoding of sarcasm features as an example, the state of a single circuit is:

$$|\psi\rangle^t_{sar} = Rx\left(h^{t'}_{sar}\right)H|0\rangle \tag{25}$$

$$= Rx\left(h^t_{sar} + b^t_{sar}\right)H|0\rangle \tag{26}$$

$$= \frac{1}{\sqrt{2}}e^{-ib^t_{sar}/2}\left(e^{-ih^t_{sar}/2}|0\rangle + e^{-ih^t_{sar}/2}|1\rangle\right) \tag{27}$$

$$= e^{-ib^t_{sar}/2}e^{-ih^t_{sar}/2}|+\rangle \tag{28}$$

where:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \tag{29}$$

$$Rx(\theta) = \begin{bmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \tag{30}$$

Let $r = e^{-ib^t_{sar}/2}, s = h^t_{sar}/2$:

$$|\psi\rangle^t_{sar} = re^{-is}|+\rangle \tag{31}$$

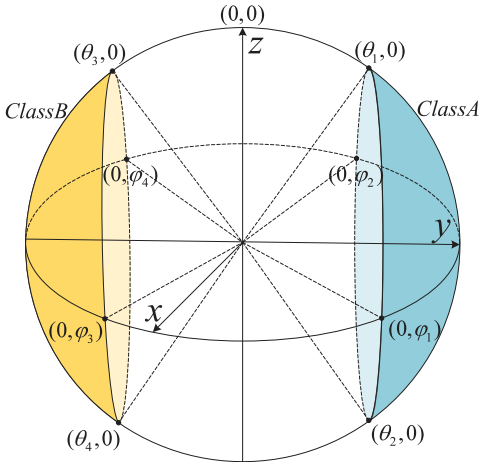


Fig. 4. Defuzzifier's Bloch sphere representation.

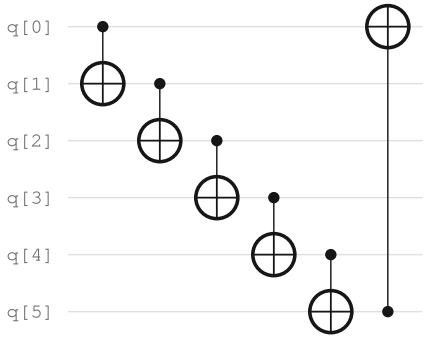


Fig. 5. Quantum entanglement circuit.

Eq. (31) obtains the sarcasm feature of complex value coding. The updated fuzzy sets are:

$$|\psi\rangle_{sar}^{t} = \left\{ \left(|0\rangle, (re^{-is})^2\right), \left(|1\rangle, (re^{-is})^2\right) \right\} \tag{32}$$

And the steps of sentiment features can be obtained in the same as of sarcasm features. From (32), it can be seen that compared with using real value membership functions, complex values have more substantial expressive power for features. For example, for sarcasm features, the membership function values corresponding to the two states re^{is_1} or re^{is_2}. Take r as the probability of occurrence, and s as the hidden attribute that has a directional effect on sarcasm detection. Therefore, the features can be better expressed by complex value representation than the real value representation.

Defuzzifier: Similar to the typical Defuzzifier process, the goal is to map the features to the corresponding categories. For a single qubit, it uses the $if-then$ rule to make the following definitions:

Definition 1. if $\theta_1 \le \theta \le \theta_2, \varphi_1 \le \varphi \le \varphi_2$, then $|\psi\rangle$ is classA
if $\theta_3 \le \theta \le \theta_4, \varphi_3 \le \varphi \le \varphi_4$, then $|\psi\rangle$ is classB
...
if $\theta_{n-1} \le \theta \le \theta_n, \varphi_{n-1} \le \varphi \le \varphi_n$, then $|\psi\rangle$ is classX.

As shown in the Bloch sphere in Fig. 4. Where θ is the angle between the state vector and the z-axis, and φ is the angle between $|\psi\rangle$ and the x-axis. θ_i and φ_i are parameters that need to be trained, i is the serial number of the parameter. A, B, X are the categories of objects. It should be noted here that A, B, X can be the same, because we do not know the distribution of these categories on the sphere, and the distribution of the same category may not be continuous. Since the radius of the Bloch sphere is 1, the coordinates in the figure are expressed in the form of (θ, φ). The graphical representation of $classA$ and $classB$ in the above $if-then$ rule is shown in the blue and yellow areas in the figure. This paper uses parameterized quantum gates to fix θ_i and φ_i. The Deffuzifier's task is to assign the fuzzy quantum state to its corresponding category. In the representation on the Bloch sphere, this means 'moving' the state vector to the interval corresponding to its category. Parameterized quantum rotation gates are used to 'move' this vector.

Let the quantum state after the Fuzzifier be $|\psi_F\rangle$, and after a layer of parameterized Rz gates, we have:

$$|\psi_z\rangle = \prod_i^{\otimes} Rz(\theta_s^i)|\psi_F\rangle \tag{33}$$

In Eq. (33), θ_s^i is the variable parameter of the Rz gate. i is the sequence number of the parameter, $i \in \{0, 1, \ldots, k-1\}$, $s \in \{sen, sar\}$.

The CNOT gate is used to perform the entanglement of the features. Six qubits are used, and the specific entanglement method is shown in Fig. 5.

Let the quantum state after entanglement to be $|\psi_E\rangle$, and after another layer of parameterized Ry gates, we have:

$$|\psi_y\rangle = \prod_i^{\otimes} Ry(\theta_s^i)|\psi_E\rangle, \qquad i \in \{k, k+1, \ldots, 2k-1\} \tag{34}$$

From Eq. (34), we obtain the transformed quantum state $|\psi_y\rangle$. The parameterized quantum gates are Rz and Ry. Because Rz and Ry gates can transform a state vector into any other state vector in Hilbert space.

Finally, all circuits are measured multiple times, and the value with the highest probability in the measurement results is taken as the predicted feature. The measurement operator $M_m = Z^{\otimes k}$ is used for the measurement. The results of the first k qubits are taken as the results of sarcasm detection, and the last k is the results of sentiment detection. They are passed through the $Softmax$ function to get the final predicted labels. Then the predicted values are put into the cross-entropy loss function, and the parameters are optimized using the Adam optimizer.

Algorithm 1 Steps of QFNN method

Input: target text features T_i^t, context text features T_j^c, target image features V_i^t, context image features V_j^c, target audio features A_i^t, context audio features A_j^c. Let N be the number of samples, n be the number of each sample's contexts, and $2k$ be the number of qubits.

Output: sarcasm label L_{sar_i}, sentiment label L_{sen_i}.

Encoder.
for $i \in [0, N-1], U \in \{T, V, A\}$ **do**
  $U_i \leftarrow Intramodal\ Attention(U_i^t, U_0^c \ldots U_{n-1}^c)$
end for
$M_i \leftarrow Intermodal\ Attention(T_i, V_i, A_i)$
Decoder.
for $i \in [0, N-1]$ **do**
  $I_{sen} \leftarrow GRU(M_i)$
  $I_{sar} \leftarrow GRU(M_i)$
  $h_{sen} \leftarrow Self\ Attention(I_{sen})$
  $h_{sar} \leftarrow Masked\ Attention(I_{sen}, I_{sar})$
end for
Quantum Fuzzy Composition.
$|\psi_F\rangle \leftarrow Fuzzifier(h_{sar}, h_{sen})$
$|\psi_y\rangle \leftarrow Defuzzifier(|\psi_F\rangle)$
$Q_i^{out} \leftarrow Measure(|\psi_y\rangle)$
Set Q_i^{sar} be the first k qubits of Q_i^{out}, Q_i^{sen} be the last k qubits of Q_i^{out}.
for $i \in [0, N-1]$ **do**
  $L_{sar_i} \leftarrow Softmax(Q_i^{sar})$
  $L_{sen_i} \leftarrow Softmax(Q_i^{sen})$
end for

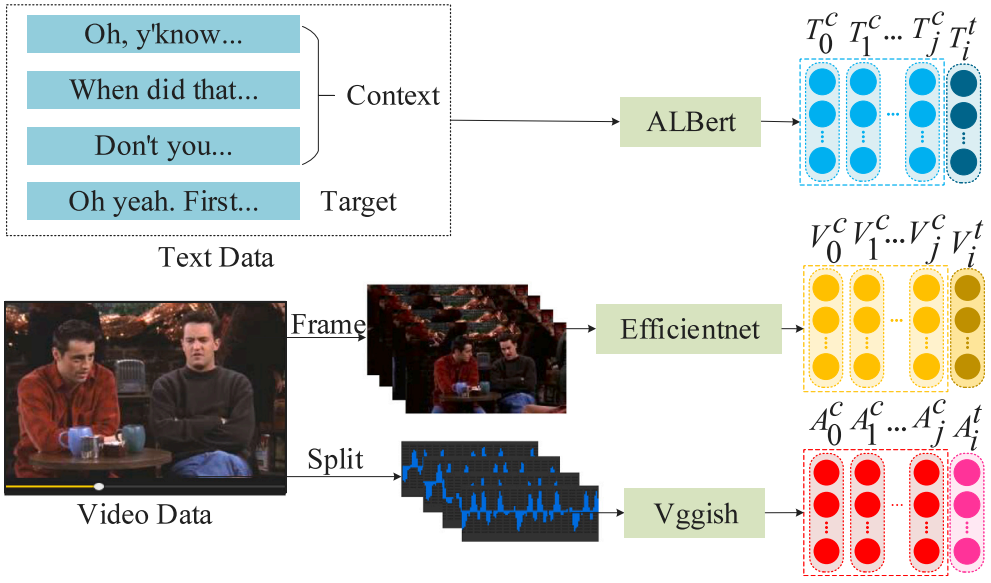


Fig. 6. Preprocessing multimodal data.

Table 1
Statistics of the datasets.

DataSet	Task	Label	NOS	Proportion (%)
Mustard	Sarcasm	TRUE	345	50.0
		FALSE	345	50.0
	Sentiment	Positive	210	30.4
		Neutral	89	12.9
		Negative	391	56.7
Memotion	Sarcasm	TRUE	5448	77.9
		FALSE	1554	22.1
	Sentiment	Positive	631	9.0
		Neutral	2201	31.5
		Negative	4160	59.5

3.3. Training procedure

First, the raw information needs to be pre-processed. As shown in Fig. 6, FFmpeg is used to separate images and audio information from videos. The images are obtained by frame extraction from the videos. To make full use of the visual information in the videos, evenly frames are extracted from each video—dividing the video into several segments of the same duration and extracting the first frame of each segment. For audio information, a sampling rate of 16 kHz is used. For the separated raw information, pre-trained models are used for feature extraction. Albert [46] is used for text feature extraction, EfficientNet [47] for image feature extraction, and Vggish [48] for audio feature extraction.

After the raw information is pre-processed, multimodal features with context are obtained. The following algorithm process is shown in Algorithm 1. In the encoding process, this paper refers to the method in [22], using Intra-modal Attention to aggregate context features under the same modality and Inter-modal Attention to fuse multimodal information. In decoding, GRU and Masked-attention are used to decode the fusion features into Sarcasm and Sentiment features. Then, the expressiveness of the features is enhanced through the Fuzzifier. The class of the features is identified by the Defuzzifier. Finally, the circuit is measured, and the final prediction value is obtained through the activation function.

4. Experiments

The primary aim of the experiment is to test the performance of the model and find answers to the following questions:

1. Is QFNN's performance satisfactory?
2. What is the effect of integrating information from different modalities on QFNN?
3. How does the order of multi-task affect the model?
4. Does QFNN already have sufficient expressive power?
5. Is QFNN robust enough against noise?

4.1. Experimental settings

4.1.1. Experimental environment

The experiment was conducted on the x86 platform, using GPU to simulate the classical network and quantum circuit. PyTorch is used to build the classical network, and the Torch Quantum library [49] is used to build the quantum circuit and simulate the quantum environment. The cross-entropy loss function is used as the loss function.

4.1.2. Datasets

Mustard [50] and **Memotion** [51] datasets are used for the proposed model evaluation. Both datasets are multimodal and have both sarcasm and sentiment labels. The size and labels of the specific datasets are shown in Table 1. Mustard includes video clips of character dialogues from TV series like *The Big Bang Theory and Friends*, as well as corresponding text information. Mustard dataset has 2 sarcasm labels and 3 sentiment labels. And the size of Mustard's training sets is 690. The Memotion dataset includes 6992 training sets, consisting of text and images. The content is memo images from social media. The number of sarcasm labels for Memotion is 2 and the number of sentiment labels is 3.

4.1.3. Hyperparameter settings

The hyperparameters in this experiment encompass the relevant parameters of the quantum circuit. The hyperparameter settings in the network are shown in Table 2.

Table 2
Hyperparameter settings.

Hyper parameter	Mustard	Memotion
Text features (dims.)	(4,24,768)	(4,24,768)
Image features (dims.)	(4,56,56)	(4,56,56)
Audio features (dims.)	(4,128)	/
Batch size	500	1000
Learning rate	0.01	0.03
Optimizer	Adam	Adam
Quantum bits	6	6
Activation Func	$Softmax$	$Softmax$
Measurement	Z Gate	Z Gate
Quantum encoding gate	Rx Gate	Rx Gate

4.1.4. Evaluation metrics

For binary classiciation tasks, we used $Precision$, $Recall$, $F1$, and $Accuracy$ to evaluate model performance.

$$Precision = \frac{TP}{TP + FP} \tag{35}$$

$$Recall = \frac{TP}{TP + FN} \tag{36}$$

$$F1 = \frac{2Precision \times Recall}{Precision + Recall} \tag{37}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FN + FP} \tag{38}$$

Where TP is the number of correctly classified positive samples; TN is the number of correctly classified negative samples; FP is the number of incorrectly classified positive samples; FN is the number of incorrectly classified negative samples.

For multiclass classification tasks, we used $Macro-Precision(MP)$, $Macro-Recall(MR)$, $Micro-F1$, and $Accuracy$. Suppose the categories in the classification task are $t \in \{a, b, c\}$, so we first calculate the $Precision$ and $Recall$ for each category. Then, calculate the average of the computed $Precision$ and $Recall$:

$$Macro - Precision = \frac{P_a + P_b + P_c}{3} \tag{39}$$

$$Macro - Recall = \frac{R_a + R_b + R_c}{3} \tag{40}$$

Here P and R represent the $Precision$ and $Recall$ of each category, respectively. Then, $Micro - F1$ can be calculated:

$$F1 = \frac{2 \times MP \times MR}{MP + MR} \tag{41}$$

For $Accuracy$, since this is a global indicator, the computation for multiclass classification is similar to binary classification. The denominator is still the total number of samples, and the numerator is the number of correctly predicted samples.

4.1.5. Baseline methods

The baselines for the experiment comparison are as follows:

CNN: In convolutional neural network (CNN), the dimension of the original multimodal features was reduced through convolution and pooling operations. Then, the features were combined and divided into two tasks for further convolution, and the $Softmax$ function was applied to obtain the final result. This model did not use contextual information.

SVM: Support vector machine (SVM) is used for classification, and the text features were extracted using Albert. Two groups of tests were conducted, one group only detected the target features, and the other group SVM (C) combined with the context features for detection.

RCNN-RoBERTa (R-RB) [52]: RoBERTa model is used for feature extraction, and further used a BiLSTM layer, an attention layer, and a pooling layer. The $Softmax$ function was employed for the final result. This model only used text information.

UPB-MTL (UM) [53]: ALBERT model waa used to extract text features, and VGG-16 to extract image features. An attention mechanism was employed to explore the connection between the two modalities and the hidden features in the feature vector to achieve final output. This model did not use context features.

QCNN [30]: This model used the structure of QCNN in the QNN. The Decoder and Encoder part is the same as our model. But QCNN has no Fuzzifier or Defuzzifier.

4.2. Performance analysis

4.2.1. Comparison of different algorithms

As shown in Table 3, QFNN has reached the level of classical networks in terms of performance under different circuimstances on the Mustard and Memotion datasets. In the Mustard dataset, the F1 value of the QFNN sarcasm detection task reached 68.88%, which is 1.33% higher than RCNN-RoBERTa and 5.05% higher than UPB-MTL. In sentiment detection, the F1 of QFNN reached 47.76%, which is 2.43% higher than RCNN-RoBERTa and 4.71% higher than UPB-MTL. This is due to the superior ability of QFNN to aggregate contextual features and multimodal features.

In the Memotion dataset, the performance of QFNN is not as good as on the Musatard dataset. In the sarcasm detection task, the F1 of QFNN reached 51.77%, which is 1.38% higher than RCNN-RoBERTa and 0.66% higher than UPB-MTL.

As shown in Fig. 7, in the sarcasm detection tasks on the two datasets, QFNN has already surpassed others. The same is true for sentiment detection tasks. In the sentiment detection case on the Memotion dataset, QFNN achieves better performance in terms of accuracy than other models.

We also compared QFNN with QNN model. In the sarcasm detection task on the Mustard dataset, the accuracy of QFNN is 23.32% higher than QCNN. In the sentiment detection task, QFNN's accuracy is 7.78% higher than QCNN. In the sarcasm detection task on the Memotion dataset, QFNN's accuracy is 13.31% higher than that of QCNN. In the sentiment detection task, QFNN's accuracy is 5.87% higher than that of QCNN. It is because, compared to our model, QCNN has no complex-valued representation of the features and fuzzy logic support, which lead to poor representation of sarcasm and sentiment features. This shows that adding fuzzy logic to QNN can enhance the expression of features.

4.2.2. Effect of different data modality

Table 4 and Fig. 8 show the comparison of model performance using different modalities. It can be seen that for sarcasm detection, the accuracy reached 68.88% when the three modalities were used simultaneously. However, the accuracy was only 41.87% when only image and audio information were used. This performance indicates that text information is the most important for sarcasm detection. On the other hand, the accuracy of using text and audio information is 3% higher than that of text and images, which shows that audio, or the tone of human speech in this dataset, also plays a certain role in sarcasm detection. It can be found that using visual modality information model does not performs well compared to the other two modalities. That is because, in visual modality information processing, the frame extraction method is used. Human expressions are continuous, and rough frame extraction may ignore the subtle changes in human expressions and lose some information. On the other hand, in sentiment detection, the performance of the model trained with text and image information is quite good. This shows that in sarcasm detection, the change in images or human facial expressions does not help the model's recognition much. In addition, for a single modality, it was found that the accuracy of sarcasm detection using text information was 19.98% higher than that using image and 22.64% higher than that using audio information. This means that sarcasm factors in textual information are easier to detect than in image and audio information.

For sentiment detection, the performance is optimal under the two modalities of text and image, with an accuracy of 56.03%. The accuracy

Table 3
Comparison of different models on Mustard and Memotion datasets.

DataSet	Method	Sarcasm				Sentiment			
		Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
Mustard	CNN	55.32	55.34	55.31	55.32	40.01	40.01	40.00	40.00
	SVM	63.21	63.21	63.21	63.21	41.45	42.56	42.55	42.55
	SVM (context)	63.52	63.50	63.55	63.52	42.54	42.45	42.40	42.42
	RCNN-RoBERTa	66.47	68.67	66.46	67.55	45.33	45.33	45.33	45.33
	UPB-MTL	63.82	64.33	63.33	63.83	43.21	42.35	43.78	43.05
	QCNN	45.56	45.47	45.55	45.51	40.00	40.02	40.00	40.01
	QFNN	68.88	68.89	68.88	68.88	47.78	47.75	47.78	47.76
Memotion	CNN	44.11	44.13	44.12	44.12	35.05	35.13	35.05	35.09
	SVM	45.94	46.13	45.88	46.00	34.96	35.01	35.00	35.00
	SVM (context)	46.63	46.67	46.66	46.66	35.15	35.16	35.15	35.15
	RCNN-RoBERTa	49.99	50.44	50.35	50.39	35.54	35.65	35.46	35.55
	UPB-MTL	51.03	50.88	51.35	51.11	35.31	35.33	35.32	35.32
	QCNN	38.46	38.40	38.43	38.41	36.33	36.34	36.33	36.33
	QFNN	51.77	51.67	51.88	51.77	42.20	35.66	35.58	35.62

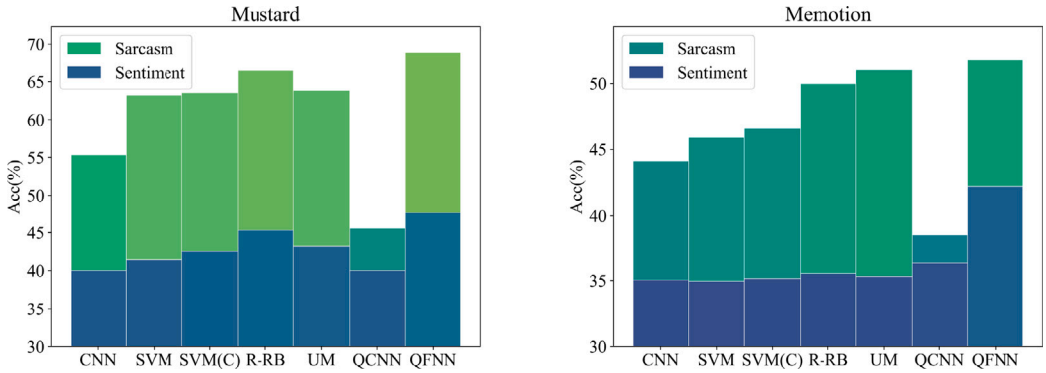


Fig. 7. Performance comparison on mustard and memotion.

Table 4
QFNN performance using different data Modalitiy on Mustard dataset.

Modals	Sarcasm		Sentiment	
	F1	Acc	F1	Acc
Text+Image+Audio (TIA)	68.88	68.88	47.76	47.78
Text+Image (TI)	48.13	47.88	55.47	56.03
Text+Audio (TA)	50.86	50.88	32.33	31.94
Image+Audio (IA)	42.13	41.87	30.14	30.15
Text (T)	52.66	52.65	33.33	33.33
Image (I)	32.66	32.67	51.33	51.33
Audio (A)	30.02	30.01	31.51	31.50

Table 5
Performance on different multi-task orders.

DataSet	Task sequence	Sarcasm		Sentiment	
		F1	Acc	F1	Acc
Mustard	sarcasm→sentiment	52.13	51.97	10.98	11.11
	sentiment→sarcasm	68.88	68.88	47.76	47.78
Memotion	sarcasm→sentiment	43.13	42.88	24.00	23.87
	sentiment→sarcasm	51.77	51.77	42.20	42.20

Table 6
Performance on different quantum circuits.

DataSet	Models	Sarcasm		Sentiment	
		F1	Acc	F1	Acc
Mustard	With Multitask CNOT Gate	56.66	56.65	45.56	45.56
	No Bias	46.67	46.67	41.11	41.11
	QFNN	68.88	68.88	47.76	47.78
Memotion	With Multitask CNOT Gate	45.44	45.44	32.53	32.53
	No Bias	36.54	36.55	31.13	31.11
	QFNN	51.77	51.77	42.20	42.20

of using three modalities, and using image and audio information is not high, as using text and image. This shows that in sentiment detection, human facial expressions are the most important factor. In addition, the accuracy of using three modalities is 8.25% lower than using text and image. It can be found that when audio information is added, the overall performance of the model decreases. This is due to the complexity of human tone exceeding the limit that the training model can express. Therefore, when the model cannot fully analyze the features of the tone, using this modality only mislead the model to make wrong predictions. In addition, for a single modality, it was found that the accuracy of sentiment detection using image information was 18% higher than using text, and 19.83% higher than using audio. This means that sentiment factors in image are easier to detect than in text and audio information.

4.2.3. Effect of tasks' order

Table 5 shows the performance of the model when predicting different task orders in multi-task learning setting. On the Mustard dataset, when performing sarcasm detection, the accuracy of sentiment→sarcasm is 16.91% higher than the accuracy of sarcasm→sentiment. The sentiment detection is 36.67% higher. On the Memotion dataset, the former is 8.89% higher than the latter in sarcasm detection, and sentiment detection is 18.33% higher. These results show that there are hidden factors in the results of sentiment detection that can affect sarcasm detection. On the contrary, the results of sarcasm detection do not help much in sentiment detection. This shows that the complexity of sarcasm factors in human language is higher than that of sentiment factors and is more difficult to capture.

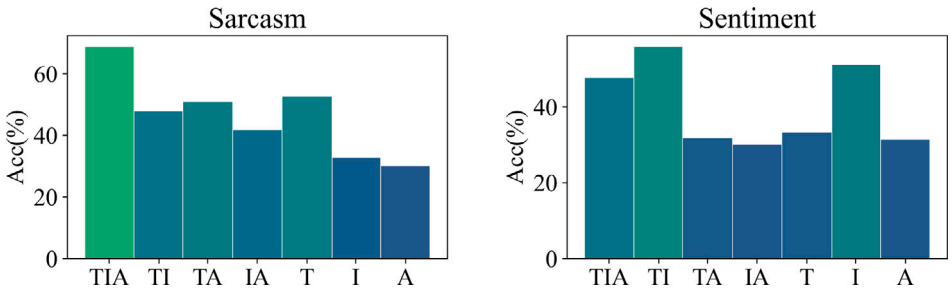


Fig. 8. Comparison of performance under different modalities.

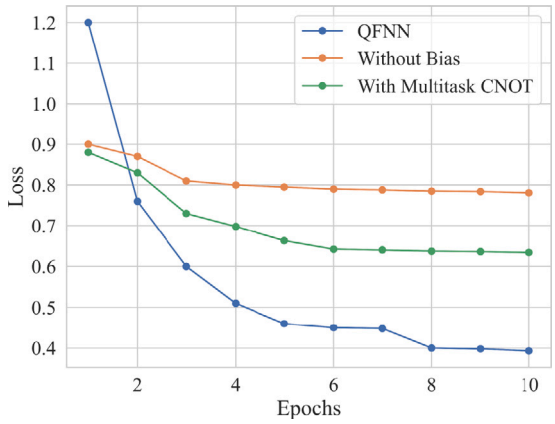


Fig. 9. Gradient descent of different quantum circuits.

4.3. Quantum circuit analysis

In Table 6, we mainly aim to uncover the balance point of the expressive power and complexity of the quantum circuit. The complexity of the circuit increases as the depth of the circuit and the number of qubits increase. This causes the parameters' gradient of the QNN to converge to 0 during training. This phenomenon, which is similar to the disappearance of gradients in classical neural networks, is known as the barren plateau. When the complexity of the circuit increases, the expressive power also increase, but this bring the barren plateau phenomenon, making the model unable to continue training [54]. Therefore, we conducted three controlled experiments. In the first experiment, CNOT gates were added into QFNN to associate the features of the two tasks. In the second experiment, the bias of features was removed in QFNN. In the third experiment, QFNN had no changes.

We tried to associate the features of the two tasks in the quantum circuit to enhance the connection between the tasks. We want to figure out whether enhancing the connection between two tasks in the circuit can make the model perform better. The qubits of the two tasks are associated using the CNOT gate. Compared with QFNN on the Mustard dataset, the accuracy of sarcasm detection of QFNN with multitask CNOT gate was 12.22% low. And in sentiment detection, the accuracy of QFNN with multitask CNOT gate was 2.22% lower than QFNN. Adding such an association did not make the network perform better but instead increased the overall complexity of the network. Fig. 9 shows that after 10 epochs, the gradient descent of QFNN with the association between the two tasks became extremely slow, which means the barren plateau phenomenon occurred. This also led to the underfitting of the entire network. Therefore, the association between two tasks in the circuit does not make the model perform better but increases the complexity of the circuit.

Adding more components gives the model higher expressive power, but it leads to the barren plateau phenomenon. Therefore, we tried to keep fewer components. From the Table 6, it can be seen that after removing the bias, the accuracy of sarcasm detection on the Mustard dataset dropped by 22.21%, and sentiment detection dropped by 6.67%. On the Memotion dataset, the accuracy of sarcasm detection dropped by 15.23%, and sentiment detection dropped by 11.09%. After removing the bias, the expression of complex values is also lost. Due to the lack of expressive power, the model cannot fit the training data, so a phenomenon similar to the barren plateau appears. However, the difficulty of gradient descent is not caused by the increase in circuit complexity but due to the lack of expressive power.

4.4. Quantum encoding methods

There are three main types of well known quantum encoding methods: basic encoding, angle encoding, and amplitude encoding [55]. *Basic encoding* converts features into binary first, and then directly encodes the binary features into quantum circuits. For a binary 0, the circuit will not change; for a binary 1, an X gate will be added to the circuit. *Angle encoding* inputs the feature value into the quantum circuit as the rotation angle value of rotation gates. Common rotation gates include Rx, Rz, and Ry gates. *Amplitude encoding* first performs unitary operation on the features and then inputs the feature values into the quantum circuit as the amplitude of the ground state.

This experiment tested the performance of three encoding methods respectively, and the results are shown in Table 7. On Mustard datasets, the sarcasm detection accuracy using angle encoding is 5.44% higher than using basic encoding, and 15.65% higher than using amplitude encoding. In sentiment detection, the accuracy of using angle encoding is 5.45% higher than basic encoding, and 14.66% higher than amplitude encoding. On Memotion datasets, the sarcasm detection accuracy using angle encoding is 4.64% higher than basic encoding, and 15.54% higher than amplitude encoding. In sentiment detection, the accuracy of angle encoding is 8.92% higher than basic encoding, and 5.07% higher than amplitude encoding. In general, it can be found that angle encoding has the highest accuracy in both sarcasm and sentiment detection on both datasets. Due to the excessive number of qubits required for basic encoding, it increases the complexity of the circuit and creates a barren plateau. This makes it difficult to train the model, so the model's performance deteriorate. Although amplitude encoding can reduce the number of qubits, it also causes the loss of information. QFNN uses 6 qubits, so amplitude encoding can only reduce 2 qubits. This has little impact on the complexity of the model. On the contrary, reducing the number of qubits lead to a loss of expressibility.

4.5. Quantum circuit robustness against noise

To analyze the robustness of the quantum circuit, the most important aspect is to analyze the noise resistance of the circuit. Noise has a great impact on quantum circuits. Quantum circuits with weak noise resistance can reduce the performance of QNN. Therefore, a quantum circuit with good robustness needs to have good noise resistance performance.

We simulated four typical single-qubit noises, which are applied to single-qubit gates with a certain probability, respectively:

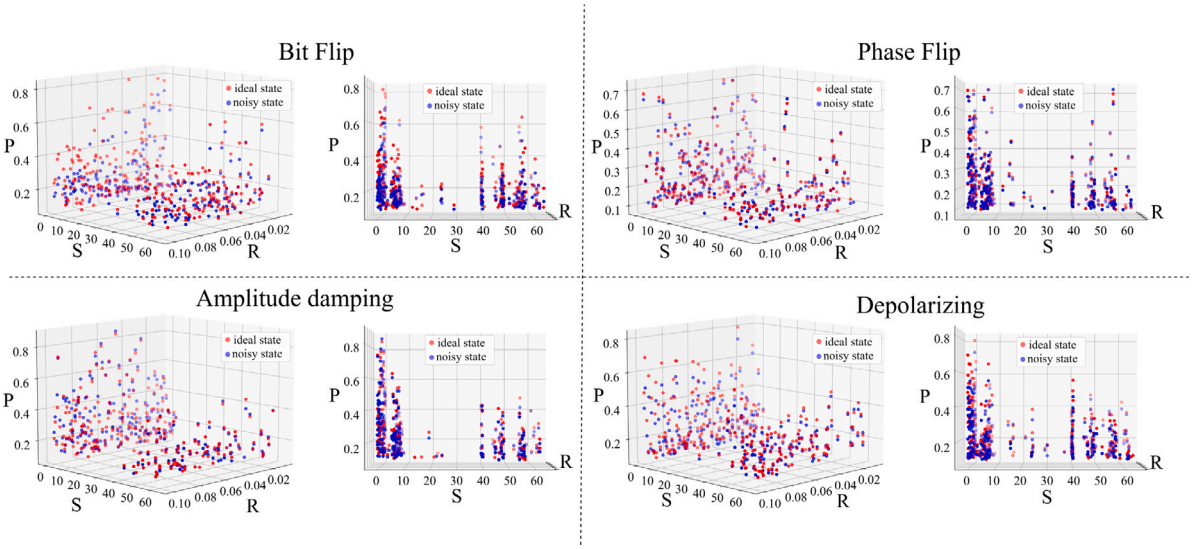


Fig. 10. Comparison of final states distribution of noisy and ideal quantum circuits. R is the probability of noise, S is the measured value, and P is the probability that the corresponding value is measured. The measured value S is the decimal representation of the measured binary number.

Table 7
Performance comparison using different quantum encoding methods.

DataSet	Encoding method	Sarcasm		Sentiment	
		F1	Acc	F1	Acc
Mustard	Basic encoding	63.45	63.44	42.34	42.33
	Amplitude encoding	53.23	53.23	33.13	33.12
	Angle encoding	68.88	68.88	47.76	47.78
Memotion	Basic encoding	47.13	47.13	33.33	33.28
	Amplitude encoding	36.23	36.23	37.15	37.13
	Angle encoding	51.77	51.77	42.20	42.20

Table 8
The fidelity of noisy quantum circuit.

p	BF	PF	AD	DP
0.01	0.97	0.98	0.98	0.98
0.1	0.83	0.98	0.97	0.92

Bit Flip (BF):

$$K_0 = \sqrt{1-p}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad K_1 = \sqrt{p}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{42}$$

Phase Flip (PF):

$$K_0 = \sqrt{1-p}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad K_1 = \sqrt{p}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{43}$$

Amplitude Damping (AD):

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix} \quad K_1 = \sqrt{p}\begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix} \tag{44}$$

Depolarizing (DP):

$$K_0 = \sqrt{1-p}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad K_1 = \sqrt{p/3}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
$$K_2 = \sqrt{p/3}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad K_3 = \sqrt{p/3}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{45}$$

The K in the above formula is the Kraus matrices, and p ($0 < p < 1$) is the probability of the corresponding operator occurring on the single-qubit gate.

For noisy quantum circuits, *Fidelity* is used to quantitatively evaluate the noise resistance of quantum circuits. Fidelity is an indicator to measure the similarity of two quantum states. Therefore, as long as the fidelity of the final state of the circuit under the noisy state and the final state of the circuit under the ideal state are calculated, the noise resistance of the quantum circuit can be evaluated. The higher the fidelity, the higher the noise resistance of the quantum circuit. The formula of fidelity is shown in Eq. (46). ρ and σ are the density matrix representations of two different quantum states.

$$F(\rho,\sigma) = Tr\left(\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\right)^2 \tag{46}$$

Four kinds of noise are simulated, and the probability of noise ranges from 0.01 to 0.1. Under the four kinds of noise, the state of the circuit is shown in Fig. 10. The fidelity of the circuit is shown in Table 8.

The noise resistence of a circuit can be accessed by the degree of overlap between the scatterplot of the state distribution in the noisy environment and the scatterplot of the state distribution in the ideal noiseless environment. From Fig. 10, it can be seen that the final state distribution of the designed circuit in a noisy environment is very close to the state distribution in an ideal environment. At the same time, from Table 8, it can be seen that even when the noise probability reaches 0.1, the fidelity of the circuit can still reach 0.8. Therefore, it can be said that QFNN has good robustness.

4.6. Expressibility and entanglement capability

In order to evaluate whether a quantum circuit can comprehensively express features, Sim et al. [54] proposed a method to measure the expressibility and entanglement capability of quantum circuits. They first calculated the fidelity probability distributions of the Haar state (which can be understood as a state with maximum expressive power) and the final state of the target circuit. Then the KL divergence is calculated for the two probability distributions, and the result is the expressibility value of the target circuit. The formula is expressed as follows:

$$\text{Expr} = D_{\text{KL}}(\hat{P}_{\text{QNN}}(F;\boldsymbol{\theta}) \parallel P_{\text{Haar}}(F)) \tag{47}$$

From Eq. (47), $\hat{P}_{\text{QNN}}$ is the fidelity probability distribution of the final state of the target quantum circuit, and P_{Haar} is the fidelity probability distribution of the Haar state. D_{KL} stands for KL divergence. F is fidelity and θ is the multiple random parameters of the target circuit. The stronger the expressibility of a circuit, the closer its target state to the Haar state, and the closer the fidelity probability distributions of the two states are. This causes the calculated KL divergence values of

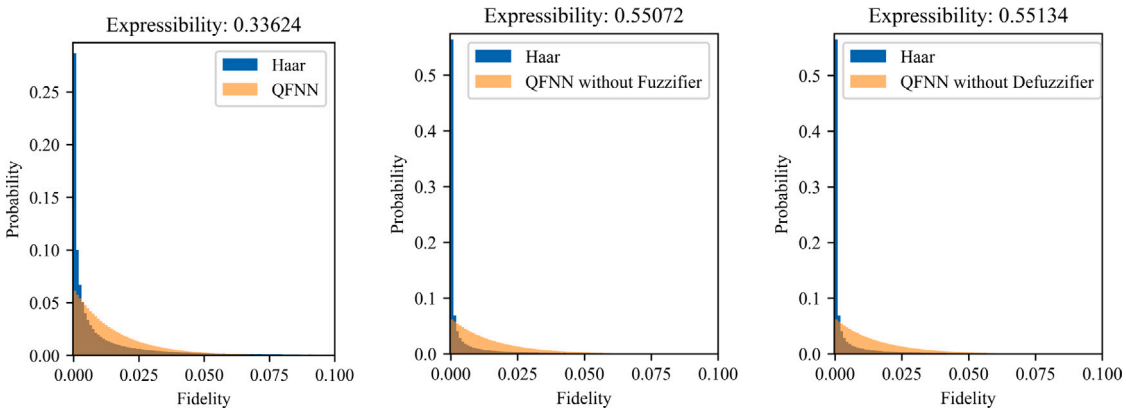


Fig. 11. Comparison of expressibility.

Table 9
Expressibility and entanglement capability comparison.

Samples	Expressibility	Entanglement capability
QFNN	0.33624	0.7518
QFNN without Fuzzifier	0.55072	0.0004
QFNN without Defuzzifier	0.55134	0.0001

the two probability distributions to approach 0. Therefore, the closer the expressibility value is to 0, the stronger the expressiveness of the circuit.

For the computation of entanglement capability, they [54] used a special measurement operator named Meyer-Wallach measure. For a circuit with strong entanglement capability, the results measured by the Meyer-Wallach measure will be larger. The process of this approach is as follows:

$$Q^{MW} = \frac{2}{|\vec{\theta}|} \sum_{\theta_i \in \vec{\theta}} \left(1 - \frac{1}{n} \sum_{k=1}^{n} Tr(\rho_k^2(\theta_i)) \right) \tag{48}$$

θ is the circuit's parameter, $|\vec{\theta}|$ is the number of test parameters, n is the number of qubits, and ρ_k is the partial trace. Through these two methods, the expressibility and entanglement capability of QFNN are evaluated. Three settings are used, first one is QFNN, second is QFNN without Fuzzifier, and third is QFNN without Defuzifier. The comparison of expressibility is shown in Fig. 11. The expressibility of QFNN is 38.95% higher than QFNN without Fuzzifier, and 39.01% higher than QFNN without Defuzifier. In addition, the fidelity probability distribution of QFNN's final state covers the most of fidelity probability distribution of Haar states. It can be concluded that QFNN has the best expression ability. This is because QFNN has more circuit layers and more quantum gates than the others.

The comparison of entanglement capabilities is shown in Table 9. The entanglement ability of QFNN is at least 0.7 higher than the other two. This is because QFNN without Fuzzifier lacks the Hadamard gates, and QFNN without Defuzifier lacks the CNOT gates between qubits. Both types of gates are essential for the entanglement of quantum circuits. Therefore, it can be said that QFNN has sufficient entanglement capabilities. Besides, the entanglement capability value of QFNN reaches 0.7518. This value is not very high, but it is enough for our circuit. Because too much entanglement can increase the complexity of the circuit, and it can have a negative effect on the model.

4.7. Hyper-parameter analysis

To understand the impact of hyperparameters tuning on training performance, this study analyzed hyperparameter sensitivity. Specifically, the learning rate (LR), training epochs (TE), batch size (BS), qubits number (QN), and activation function (AF) were chosen for testing. In each test, we only tuned one of these parameters, and compared the degree of average change of each hyperparameter. During the experiment, only three settings were tested when the sensitivity of qubits number was analyzed: a circuit with 4 qubits, a circuit with 5 qubits, and a circuit with 6 qubits. This is because a very small qubits number cannot form a QNN, and if it is too large, training becomes infeasible.

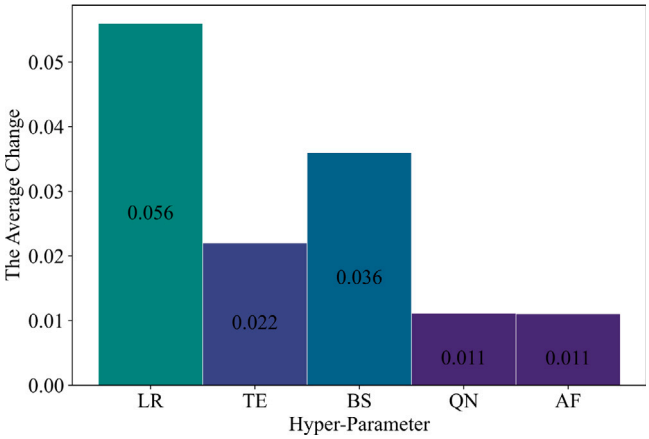


Fig. 12. Hyperparameter analysis.

Fig. 12 shows the degree of average change for each hyperparameter. It can be found that the change in the learning rate cause the most significant impact on the QFNN, with the degree of average change reaching 5.6%. The next influential factor is the batch size, with a degree of average change of 3.6%. Following that is the training epoch, which had a degree of average change of 2.2%. The least impact on model performance came from the qubits number and the activation function, with only a 1.1% degree of average change. Therefore, more attention should be paid to adjusting the learning rate and the batch size.

5. Conclusion and future work

This paper proposes QFNN, a multi-modal fusion and multi-task learning algorithm with a Seq2Seq structure based on classical and quantum networks. For multi-modal features, Intra-modal Attention is used to aggregate the context of unimodal features, and Inter-modal Attention is used to fuse multimodal features. Furthermore, we combine the fuzzy system and QNN to enhance the expressiveness of sentiment and sarcastic features. Specifically, the Fuzzifier realizes the complex value fuzzy encoding of the features into QNN. Then, the features are Defuzzified through the *if-then* rule and the parameterized quantum

circuit to get the output results. The results show the QFNN outperform several methods in sarcasm and sentiment detection tasks on the Mustard and Memotion datasets. In addition, the fidelity test under the noisy environment, QFNN has also excellent robustness against noise.

There are several challenges with QFNN. Firstly, due to the limited quantum resources of the current hardware for simulating quantum computation, QFNN may not have a significant impact on larger datasets. When the hardware conditions improve in the future, the circuit scale of QFNN can be expanded to further improve the performance of sarcasm and sentiment detection tasks. Additionally, while this research mainly focuses on sarcasm and sentiment detection, QFNN remains unexplored in other domains. There is potential for its application in various fields, such as humor detection, fake news detection, intelligent transportation, intelligent medical care, etc. Therefore, we are committed to exploring the application of QFNN in other fields. As quantum computers continue to mature, the practical feasibility of QFNN can be further validated by implementing it in a real quantum computer environment.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data is available at https://github.com/prayagtiwari/QFNN.

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